

Capturing Provenance of Academy Examples with Flowcept

A 1-hour, hands-on tutorial — CPU-friendly, Argo/OpenAI/local LLM

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What we'll do

Take the **stock Academy examples**, instrument each with **Flowcept** to capture provenance (**zero changes to agent logic**), and **inspect** what happened — including **where things fail**.

Part	Content
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Terminal-only (no images): text summaries, ASCII dashboards, a markdown provenance card — runs over SSH on Aurora. Every run writes to its own `runs/<id>_<date-time>/`.

How the tutorial is laid out

Eight stock Academy examples, each a **step-by-step exercise**. As shipped each one *runs but records nothing*; you turn provenance on one **STEP** at a time by uncommenting a block and re-running.

```
exercises/  
  local/      run on a laptop / login shell    (python exercise.py)  
  aurora/     run on ALCF Aurora                (qsub submit.pbs; shared env.sh)  
    <id>/     exercise.py  solution.py  README.md  [submit.pbs]
```

- `exercise.py` — STEP 0 baseline, then STEP 1 capture, 2 inspect, 3 analyze, 4 card, 5 query (uncomment as you go)
- `solution.py` — every step already enabled (the reference)
- `flowcept_academy/` is a small **library** the exercises build on

Setup — one command, one env, from scratch

```
bash setup/install.sh          # builds ONE conda env for all 8 (CPU)
conda activate flowcept-academy
cd exercises/local/01-actor-client && python exercise.py
```

install.sh builds a single flowcept-academy conda env and installs Flowcept (agentic branch) + Academy + Parsl + LangGraph + a CPU build of torch/transformers, plus rdkit/xtb-python for example 07's real chemistry, and drops the offline Flowcept settings in place. **One env for the whole tutorial** — no venv, no per-exercise env. No Redis, no Mongo, no GPU.

Why provenance?

Why capture provenance at all?

Provenance = the recorded history of *what ran, on what inputs, producing what outputs, when, where, and how it all connects*.

- **Reproducibility** — rerun and get the same result; know exactly what code, data, and parameters produced a number.
- **Debugging** — when something breaks, see *which* step failed, on *what* input, and why — not just a stack trace.
- **Trust & audit** — verify and defend results; a scientific claim is only as good as the record behind it.
- **Cost & performance** — see where time and tokens go.

For classic HPC workflows this is valuable. For **agentic** systems it becomes **essential** — next slide.

Why it matters more for agentic systems

Traditional code does what you wrote. **Agents decide what to do at runtime** — so the “what happened” isn’t in your source.

- **Non-determinism.** LLM calls sample — the same prompt yields different outputs. You *cannot* reproduce a run without recording the actual prompt, response, model, and tokens.
- **Autonomy & emergence.** @loops act on their own; agents choose which tool/peer to call. The control flow is *decided*, not declared — capture is the only way to know the real path taken.
- **Distribution across boundaries.** Work spans agents, processes, nodes, and *frameworks* (Academy + LangGraph + Parsl). Lineage that crosses those boundaries is invisible without provenance.
- **Opaque reasoning.** Prompts, tool calls, and delegations bury the “why.” Provenance keeps every prompt/response — the audit trail.
- **Failure is normal.** Small LLMs propose malformed inputs; agents derail mid-plan. You need the exact failing step + input.

Part 1 — Academy, a reminder

A framework for **distributed multi-agent systems** for science.

- `Agent` — a class holding state + behaviors
- `@action` — a callable behavior (request/response)
- `@loop` — autonomous background behavior
- `Handle` — a remote reference to an agent
- `Manager` + `Exchange` — launch agents & route messages

Scales from local threads → processes → Globus Compute / Parsl across nodes.

Academy: the shape of an agent

```
class Coordinator(Agent):
    @action
    async def process(self, text: str) -> str:
        text = await self.lowerer.lower(text)           # cross-agent call
        return await self.reverser.reverse(text)        # cross-agent call

async with await Manager.from_exchange_factory(
    factory=LocalExchangeFactory(),
    executors=ThreadPoolExecutor()) as manager:
    coord = await manager.launch(Coordinator, args=(lo, re))
    await coord.process("DEADBEEF")                    # -> "feebdaed"
```

These are Academy's own examples (github.com/academy-agents/academy); we instrument them unchanged.

Part 2 — Flowcept + examples 1, 2, 3

Flowcept: provenance, one line to capture

Flowcept records *what ran, with what inputs/outputs, when, and how it connects* — the provenance graph. Its *agentic* plugin patches Academy's Runtime, so capture is automatic.

Model: campaign → workflow → task (action / loop / llm_call), with used/generated and parent_task_id lineage.

Zero agent-code changes

Wrap the run; the plugin does the rest.

Turn capture on — STEP 1

```
from flowcept_academy.capture import captured      # zero agent-code changes
from flowcept_academy.util import capture_run      # -> runs/<id>_<date-time>/

with capture_run("03-agent-agent") as run_dir:
    with captured(workflow_name="03-agent-agent"):
        result = asyncio.run(run())               # the stock example, now captured
        # -> flowcept_buffer.jsonl in run_dir (offline: no Redis / no Mongo)
```

In the exercise you just uncomment this block and re-run. Everything a run produces — buffer, perf CSV, the markdown card — lands together in the run dir:

```
cd exercises/local/03-agent-agent && python exercise.py
# -> runs/03-agent-agent_<date-time>/ : flowcept_buffer.jsonl + card.md + perf
    .csv
```

Outcomes: examples 1, 2, 3

Each example exercises a different slice of the provenance model:

Example	Provenance captured
01 actor-client	agent lifecycle + actions (increment → count)
02 agent-loop	autonomous @loop start/exit events (group_id)
03 agent-agent	cross-agent calls (caller → callee.action)

e.g. 03 shows the chain DEADBEEF → deadbeef → feebdaed across three agents.

How to inspect the provenance

Three built-in ways, for *any* run — all terminal (from an example folder):

```
# 1) tailored, content-aware analysis + text dashboard (printed by each run)
python solution.py                                # STEPs 2-4 print the report

# 2) analyze any captured buffer / whole run dir
python ../../../../provenance/analyze.py runs/03-agent-agent_*

# 3) interactive queries: ask(...) turns NL into pandas
python ../../../../provenance/query.py runs/03-agent-agent_* --ask "show the
    cross-agent calls"
```

The analysis auto-adapts: it reports only what the example produced (actions, loops, cross-agent calls, LLM calls, ...) — and any failures. Flowcept's markdown provenance card is written into each run dir.

The provenance dashboard — in the terminal

No images: each run prints an ASCII dashboard (records by subtype; LLM tokens by agent, or tasks by activity; capture overhead). Real 06-11m run:

```
=====
06-11m -- provenance
=====

records by subtype:
  academy_lifecycle | ##### 4
  llm_call           | ##### 2
  academy_action     | ##### 2
LLM tokens by agent:
  Orchestrator       | ##### 140
capture overhead (total ms by category):
  flush              | ##### 1.135
  llm_hook            | ##### 0.892
  intercept_task      | ##### 0.836
=====
```


Part 3 — Advanced: agents & LLMs

LLM backend: Argo → OpenAI → local CPU model

- `ARGO_USER` set → **ANL Argo gateway** (on Aurora / ANL network), native tool calling
- else `OPENAI_API_KEY` set → **OpenAI** (`api.openai.com`, `OPENAI_MODEL`, default `gpt-4o-mini`), native tool calling
- else → **local CPU model** (Qwen2.5-0.5B-Instruct, offline, no GPU)
- **no mock**: every response is from a real model; `chat()` raises rather than fabricate text

Checked in priority order, first match wins; force with `FLOWCEPT_TUTORIAL_LLM=argo|openai|local`. Every call is recorded as an `llm_call` (model, prompt, response, tokens), linked to its action.

Examples 6 & 8: LLM-driven agents (vendored upstream)

```
# 06: the upstream Orchestrator builds a LangChain ReACT agent over a  
# tool that messages a separate MySimAgent -- the LLM decides to call it.  
agent = create_agent(make_chat_model(), tools=[sim_tool])  
result = await agent.ainvoke({"messages": [("user", question)]})
```

- **06 llm:** ReACT agent → llm_call + langgraph_node/graph, and a cross-agent tool call when the LLM invokes the tool
- **08 discussion:** multi-agent group chat — three GroupChatAgents (Manager, Assistant, Senior Engineer) + RoundRobinGroupChatManager with a supervisor @loop; per-turn llm_calls, tokens per agent, bounded by max_rounds

The LLM is injected via make_chat_model() (Argo → OpenAI → local) exactly where upstream builds ChatOpenAI — **zero other agent-code changes.**

What gets captured — a real 06-llm run

The upstream ReACT agent, run on the **local 0.5B** model. One @action wraps one langgraph_graph/node, which drives one llm_call — all sharing the campaign:

```
* Records (subtype activity)
  academy_action      answer                <- Orchestrator.@action
  langgraph_graph     LangGraph             <- create_agent ReACT graph
  langgraph_node      model                 <- the LLM step in the graph
  llm_call             _AsyncChatHuggingFace
    prompt : ...simulated ionization energy of benzene...
    answer  : The simulated ionization energy of benzene can be
               calculated using quantum chemistry methods, such as DFT...
```

The action, the graph, its node, and the LLM call are one connected, queryable lineage. **Note:** the 0.5B model answered *directly* instead of calling the tool — provenance is exactly how you see that it skipped the tool. A tool-capable backend (Argo) instead emits a cross-agent tool_call to MySimAgent.

Part 4 — Extra advanced & failures

Example 7: cross-framework (Academy + LangGraph + real xTB)

The upstream `mol_design_agents.py`, **unchanged**: a LangGraph StateGraph (plan → tool_calling → simulate → conclude → critique → update, looping) runs inside an Academy @loop, and simulate calls **real GFN2-xTB** (RDKit → ASE → xtb).

- **one shared** campaign_id across both plugins (Academy + LangGraph)
- three layers in one lineage: academy_loop → langgraph_node → tool_call (the xTB energy)
- LLM = Argo gpt-4o (tool-capable) — drives the tool_calling node

Real run — tool_call=10, langgraph_node=9, llm_call=6, academy_loop=2 (1 campaign):

```
* nodes: plan, tool_calling(2), simulate(2), should_continue,
          conclude, critique, update
* real xTB ionization energies (charged - neutral, eV):
    simulate -> 13.1975...    simulate -> 14.1156...
* llm_call model=gpt-4o-2024-11-20 parent=tool_calling
```

Example 5: Parsl

```
class SimulationAgent(Agent):  
    @action  
    async def run_expensive_task(self) -> int:  
        return await asyncio.wrap_future(expensive_task())    # a Parsl app
```

The agent delegates to a **Parsl** task; provenance captures the dispatching action and its result. (Example 07 shows the same pattern with real chemistry — the agent's process pool runs GFN2-xTB, each emitting a `tool_call`.)

Inspecting failures — where provenance shines

Agents (and LLMs) get things wrong. In **example 07 this is not staged** — the LLM proposes candidate molecules and some simply **can't be built** by RDKit/xTB. Flowcept records each failed `tool_call` with **status=ERROR** + real `stderr`:

```
status:  FINISHED 22    ERROR 10
!  Failures captured (status=ERROR) -- what the agent actually tried:
  tool_call/compute_ionization_energy  [ERROR]
    stderr: Parse failure for C3N20
  tool_call/compute_ionization_energy  [ERROR]
    stderr: Parse failure for CF3NC(N)=O
  tool_call/compute_ionization_energy  [ERROR]
    stderr: Parse failure for C6H5NC(N)=O
... (8 xTB parse failures + 1 propagated langgraph_node)
```

You see *which* molecule the LLM invented, that xTB rejected it, and that the successful runs still gave real energies (13.20, 14.12 eV) — the debugging payoff, and the reason agentic runs need this most.

Running on Aurora (CPU, offline, terminal-only)

Each example under `exercises/aurora/<id>/` has its own `submit.pbs`; all share `exercises/aurora/env.sh` (modules + conda env + offline LLM + settings).

```
# once, on a login node (has internet): build the env + pre-cache the local
  model
bash setup/install.sh                # -> conda env 'flowcept-academy'
source exercises/aurora/env.sh      # frameworks; conda activate; HF offline;
  settings
python -c "from flowcept_academy.util import chat; print(chat('hi'))"

# then submit an exercise (edit -A <project> first)
cd exercises/aurora/06-llm && qsub submit.pbs    # -> runs/06-llm_<date-time>/
```

`env.sh` sets `HF_HUB_OFFLINE=1` + `FLOWCEPT_TUTORIAL_LLM=local`, so compute nodes need no network (or use `ARGO_USER`). Provenance stays offline; inspect with the same terminal tools — no images to shuttle around.

- **Capture:** one line (`captured()`) instruments any Academy example — zero agent-code changes.
- **Examples:** actor, loop, cross-agent, multi-process, Parsl, LLM agents, and cross-framework LangGraph + real xTB (07).
- **LLM:** Argo → OpenAI → local CPU model.
- **Inspect:** tailored analysis, text dashboards, provenance cards — all terminal, including **failures** (status=ERROR + stderr).
- **Outputs:** every run in its own `runs/<id>_<date-time>/`.

instrument → capture → inspect (even the failures)

Thank you!

`exercises/{local,aurora} · provenance/analyze.py · slides/ · upstream`

examples: `github.com/academy-agents/academy`